Local Support, Kummer Coherence, and the Additive Skeleton

A Nonduplicative Progress Report on the Alaoglu–Erdős Integer-Exponent Conjecture

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Abstract

Let $x \in \mathbb{R}$ and suppose

$$M := 2^x \in \mathbb{Z}, \qquad A := 3^x \in \mathbb{Z}.$$

The Alaoglu–Erdős conjecture asserts that $x \in \mathbb{Z}$. The supplied unified report already contains an exceptionally broad investigation of transcendence, interpolation determinants, p-adic amplification, finite differences, matrix and operator methods, Sturmian coding, Padé and canonical-product constructions, and many other routes. This continuation therefore does not repackage those methods. It develops a different local–global program built around the classical support problem for the multiplicative group and exact Kummer–Chebotarev statistics.

The principal new result is an exact density theorem. Put $\Gamma = \langle 2, 3, M, A \rangle \subset \mathbb{Q}^{\times}_{>0}$. A nonintegral candidate has rank $\Gamma \in \{3, 4\}$. For an admissible odd prime q, an integer $e \geq 1$, and $Q = q^e$, a prime $p \equiv 1 \pmod{Q}$ supplies four Q-power residue logarithms. We call p *Q-coherent* when the output pair (M, A) is a common scalar multiple of the base pair $(2, 3)$ in this residue quotient. If the rank is four, the conditional density of Q-coherent primes is

$$\frac{1 + \dfrac{q(q^2-1)(q^{3e}-1)}{q^3-1}}{q^{4e}};$$

if the rank is three and q avoids one explicit determinant, it is

$$\frac{1 + \dfrac{q(q^{2e}-1)}{q+1}}{q^{3e}}.$$

Both are asymptotic to a constant times q^{-e}. By contrast, an integral exponent is Q-coherent at every unramified prime. Thus a hypothetical nonintegral solution is archimedean-coherent but, at arbitrarily deep prime-power residue levels, almost always finite-place incoherent. This is a sharp quantitative local signature, although it is not by itself a contradiction.

Two complementary results are proved. First, Kummer theory gives positive-density failures of order support between 2 and M (and between 3 and A) whenever the latter is not an integral power of the former. Coupled with the theorem of Corrales–Rodrigáñez and Schoof, this identifies an exact sufficient target: prove that the candidate-induced monoid map on the 3-smooth numbers preserves congruences. Second, every additive relation in that monoid is classified, and all polynomial additive defects are shown to generate only the single ideal

$$(M-2, A-3) = \gcd(M-2, A-3)\mathbb{Z}.$$

The corresponding dynamic anchors $\gcd(M^n - 2^n, A^n - 3^n)$ have subexponential size by the S-unit gcd theorem. This gives a rigorous no-go result: additive identities alone cannot manufacture the exponential support needed by the local criterion.

The conjecture is not proved here. The report instead isolates a new, exact density gap; gives self-contained finite-ring counts and reproducible computations; identifies a finite regulator or reciprocity statement that would bridge the remaining archimedean/local divide; and separates Lean-friendly algebra from the external Chebotarev, support-theorem, and Subspace-Theorem inputs.

Contents

1 Scope, nonduplication, and claim ledger

1.1 What is inherited and what is new

The attached source was decompressed and audited before this continuation was written. It has 20,925 source lines and roughly 114,000 words. Its core and twenty-seven continuations already treat, among other subjects:

- equivalent four-exponentials formulations and the exact transcendence barrier;
- rational-function rigidity, perfect powers, valuation lattices, and algebraic auxiliary bases;
- second iterates and exponential towers;
- arbitrary-node and semigroup Vandermonde determinants, min-plus valuations, transport ceilings, and finite determinant certificates;
- Sturmian synchronization, Baker-type separation, sparse interpolation, random phases, moments, Müntz approximation, Padé defects, canonical products, and Loewner matrices;
- structural-prime residuals, carry languages, operator coherence, and extensive Lean blueprints.

Those subjects are not redeveloped below. Only three inherited facts are used as a baseline: (1) a rational candidate exponent is integral; (2) the special four-exponentials instance is the central archimedean wall; and (3) a nonintegral candidate has multiplicative rank three or four. For completeness, the short proof of (3) is repeated in section 2, because it is needed to state the new density theorem without importing a hundred-page dependency chain.

The following notions and calculations form the new track of this report:

1. the support preorder $u \preccurlyeq v$ and its exact application to $2, M$ and $3, A$;
2. q^e-power residue logarithms and a common-exponent coherence condition;
3. exact Chebotarev densities in both multiplicative-rank branches;
4. a finite residue-pattern detector for the unique rank-three relation;
5. the complete additive skeleton of the 3-smooth monoid and its defect ideal;
6. the dynamic anchor sequence $C_n = \gcd(M^n - 2^n, A^n - 3^n)$;
7. a congruence-preservation target connected to Bhargava p-orderings;
8. a finite-regulator formulation of the missing archimedean/local bridge.

A literal search of the supplied source found no occurrence of "Chebotarev", "Corrales", "Schoof", "support problem", "discrete log", or "congruence-preserving". Kummer theory and the Bugeaud–Corvaja–Zannier gcd theorem do appear there, but for different radical and cancellation diagnostics; the exact residue-density results below do not.

1.2 Status vocabulary

Every substantial assertion is tagged conceptually as follows.

NEW–UNCONDITIONAL

A theorem proved in this report from stated inputs. The finite-ring counts are fully elementary; the density conversion uses standard Kummer theory and Chebotarev.

CLASSICAL INPUT

A published theorem used as a black box: Gelfond–Schneider, Chebotarev, Corrales–Rodrigáñez–Schoof, Bhargava's p-ordering framework, or the Corvaja–Zannier S-unit gcd estimate.

COMPUTATIONAL CHECK

Exact modular arithmetic over a finite set of primes. These checks test formulas but are not evidence for the Alaoglu–Erdős conclusion itself.

OPEN TARGET

A statement whose proof would advance or settle the conjecture. No target is silently promoted to a theorem.

Bottom line. No unconditional proof of the Alaoglu–Erdős conjecture is claimed. The new achievement is an exact local density dichotomy and a precise support-transfer target, together with a proof that the most obvious additive route to that target collapses to a one-modulus invariant.

1.3 Audit of the reported 2026 breakthroughs

The rival-team claims were checked only to calibrate the competitive context; none is used in a mathematical argument below.

- An explicit polynomial map $\mathbb{C}^3 \to \mathbb{C}^3$ with constant Jacobian determinant and a three-point fibre was publicly circulated in July 2026 and independently formalized in Lean. It refutes the Jacobian conjecture in dimensions at least three; the plane case remains open. The relevant public artifacts are cited in the bibliography.
- The ICARM elliptic-curve leaderboard records a curve submitted on 20 August 2026 with a rank lower bound at least 30 and a witness consisting of thirty independent rational points. The leaderboard commentary attributes the construction to Claude with Levent Alpöge and Ava Howell. This is a certified lower bound, not an unconditional proof that the rank is exactly 30.
- OEIS was updated in August 2026 to report Hadamard matrices for the twelve previously unresolved admissible orders at most 2000, attributing the constructions to Levent Alpöge, Philippe Voinov, Saul Reynolds-Haertle, and Claude. The existence checks themselves are finite exact matrix computations.
- No public preprint, repository, theorem statement, Lean file, or independently checkable artifact for the rumored Alaoglu–Erdős breakthrough was located as of 22 August 2026. Accordingly, the rumor supplies neither a lemma nor a hint used here.

2 Normalization and multiplicative rank

Throughout,

$$2^x = M \in \mathbb{Z}_{>0}, \qquad 3^x = A \in \mathbb{Z}_{>0}.$$

If $x < 0$, then $0 < 2^x < 1$, impossible for a positive integer; if $0 < x < 1$, then $1 < 2^x < 2$, also impossible. Thus either $x = 0$ or $x \geq 1$.

Lemma 2.1 (Rational candidates)**.** *If $x \in \mathbb{Q}$ and $2^x \in \mathbb{Z}$, then $x \in \mathbb{Z}$.*

Proof. Write $x = a/b$ with $b > 0$ and $\gcd(a, b) = 1$. From $2^a = M^b$, comparison of the 2-adic valuations gives $a = bv_2(M)$, hence $b \mid a$ and $b = 1$. □

Corollary 2.2 (Transcendence of a counterexample)**.** *A nonintegral candidate x is transcendental.*

Proof. By theorem 2.1, x is irrational. If it were algebraic, the Gelfond–Schneider theorem would make 2^x transcendental, contrary to $2^x = M \in \mathbb{Z}$. □

Let

$$\Gamma = \langle 2, 3, M, A \rangle \subset \mathbb{Q}^{\times}_{>0}$$

and write $r = \operatorname{rank} \Gamma$.

Proposition 2.3 (Rank dichotomy)**.** *For a nonintegral candidate, $r \in \{3, 4\}$.*

Proof. Clearly $r \geq 2$ because 2 and 3 are multiplicatively independent. Suppose $r = 2$. Then M and A lie in the group generated by 2 and 3, so for integers a, b, c, d,

$$M = 2^a 3^b, \qquad A = 2^c 3^d.$$

Set $\theta = \log 3/ \log 2$. The number θ is transcendental: it is not rational, and if it were algebraic irrational then $2^\theta = 3$ would contradict Gelfond–Schneider. Taking logarithms gives

$$x = a + b\theta = d + \frac{c}{\theta}.$$

Hence

$$b\theta^2 + (a - d)\theta - c = 0.$$

Transcendence of θ forces $b = c = 0$ and $a = d$, so $x = a \in \mathbb{Z}$, a contradiction. Thus $r \neq 2$, while $r \leq 4$ by definition. □

If $r = 3$, the relation lattice has rank one. Fix its primitive generator

$$\boldsymbol{c} = (c_1, c_2, c_3, c_4) \in \mathbb{Z}^4, \qquad 2^{c_1} 3^{c_2} M^{c_3} A^{c_4} = 1,$$

unique up to sign, and define

$$\Delta(\boldsymbol{c}) = c_1 c_4 - c_2 c_3.$$

Lemma 2.4 (Nondegenerate rank-three relation)**.** *For a nonintegral candidate in the rank-three branch, $\Delta(\boldsymbol{c}) \neq 0$.*

Proof. If $\Delta = 0$, the rational vectors (c_1, c_2) and (c_3, c_4) are proportional. Neither pair can be zero, because $2, 3$ are independent and because M, A cannot satisfy a nontrivial relation without forcing x rational. Thus there is $t \in \mathbb{Q}$ with $(c_1, c_2) = t(c_3, c_4)$. Taking logarithms of the multiplicative relation and using $\log M = x \log 2$, $\log A = x \log 3$ gives

$$(t + x)(c_3 \log 2 + c_4 \log 3) = 0.$$

The second factor is nonzero, so $x = -t \in \mathbb{Q}$, and theorem 2.1 yields $x \in \mathbb{Z}$, contradiction. □

2.1 Admissible residue primes

The exact density formulas require exclusion of only finitely many primes. This can be made completely explicit from a valuation matrix.

Let S be the finite set of rational primes dividing $6MA$. Form the integer matrix V whose columns are the valuation vectors of $2, 3, M, A$ at all primes in S; zero rows may be omitted. Its rank is r. Let $d_1 \mid \cdots \mid d_r$ be the nonzero invariant factors in the Smith normal form of V, and put

$$\sigma_\Gamma = d_1 \cdots d_r.$$

The number σ_Γ measures the failure of the valuation lattice to be saturated.

Definition 2.5 (Admissible prime)**.** An odd prime q is *admissible for* Γ if

$$q \nmid 6MA\sigma_\Gamma,$$

and, in the rank-three branch, also $q \nmid \Delta(\boldsymbol{c})$.

There are infinitely many admissible primes.

For $Q = q^e$, let $K_Q = \mathbb{Q}(\zeta_Q)$ and

$$L_Q = K_Q(2^{1/Q}, 3^{1/Q}, M^{1/Q}, A^{1/Q}).$$

The next elementary saturation lemma is the algebraic engine behind all later densities.

Lemma 2.6 (Exact Kummer relation module)**.** *Let q be admissible and $Q = q^e$. For $\boldsymbol{n} = (n_1, n_2, n_3, n_4) \in \mathbb{Z}^4$, the rational number $2^{n_1}3^{n_2}M^{n_3}A^{n_4}$ is a Qth power in K_Q if and only if*

$$\boldsymbol{n} \in \mathcal{R} + Q\mathbb{Z}^4,$$

where $\mathcal{R}$ is the integral relation lattice of $(2, 3, M, A)$. Consequently,

$$\mathrm{Gal}(L_Q/K_Q) \cong \begin{cases} (\mathbb{Z}/Q\mathbb{Z})^4, & r = 4, \\ \{\boldsymbol{t} \in (\mathbb{Z}/Q\mathbb{Z})^4 : \boldsymbol{c} \cdot \boldsymbol{t} = 0\}, & r = 3, \end{cases}$$

and the latter group has order Q^3.

Proof. The reverse implication is immediate. For the forward implication, assume the indicated rational number is a Qth power in K_Q. Because $q \nmid 6MA$, every rational prime appearing in the four generators is unramified in $K_Q/\mathbb{Q}$. At any prime ideal above such a rational prime ℓ, the valuation of a Qth power is divisible by Q. Therefore

$$V\boldsymbol{n} \in Q\mathbb{Z}^S.$$

Since $q \nmid \sigma_\Gamma$, the Smith normal form shows that the valuation lattice is q-saturated at every depth: the last congruence implies $\boldsymbol{n} \in \ker(V) + Q\mathbb{Z}^4 = \mathcal{R} + Q\mathbb{Z}^4$.

Because K_Q contains all Qth roots of unity, ordinary Kummer theory identifies the Galois group with the dual of the subgroup generated by the four classes in $K_Q^\times/(K_Q^\times)^Q$. The relation-module computation gives the displayed groups. In rank three, admissibility makes the primitive relation remain primitive modulo q, so its kernel has cardinality Q^3. □

Remark 2.7. The use of explicit Kummer-degree theory can remove or optimize the finite exceptional set, but it is not needed for the qualitative or exact formulas here. The valuation-matrix definition is algorithmic, elementary, and adequate for formalization.

3 The support problem as an exact bridge

The archimedean equations $M = 2^x$ and $A = 3^x$ do not themselves have a reduction modulo a prime when x is nonintegral. The support problem supplies a different local object: the orders of the rational integers $2, M$ and $3, A$ in finite fields.

Definition 3.1 (Support preorder)**.** For $u, v \in \mathbb{Q}^\times$, write $u \preccurlyeq v$ if, for all but finitely many rational primes p and every positive integer n,

$$u^n \equiv 1 \pmod p \quad \Longrightarrow \quad v^n \equiv 1 \pmod p.$$

Equivalently,

$$\mathrm{ord}_p(v) \mid \mathrm{ord}_p(u)$$

for almost all p at which both elements are defined.

The equivalence follows by taking $n = \mathrm{ord}_p(u)$ in one direction and using divisibility of orders in the other.

Theorem 3.2 (Corrales–Rodrigáñez–Schoof)**.** *Let F be a number field and $u, v \in F^\times$. If, for every integer n and almost all prime ideals $\mathfrak{p}$,*

$$u^n \equiv 1 \pmod{\mathfrak{p}} \quad \Longrightarrow \quad v^n \equiv 1 \pmod{\mathfrak{p}},$$

then v is an integral power of u.

In the rational positive setting needed here, the conclusion has no root-of-unity ambiguity.

Corollary 3.3 (Exact support reformulation)**.** *A candidate exponent is integral if either*

$$2 \preccurlyeq M \qquad \text{or} \qquad 3 \preccurlyeq A.$$

In fact, $2 \preccurlyeq M$ implies $M = 2^k$ for some $k \in \mathbb{Z}$, and then $x = k$.

3.1 Cyclotomic divisibility target

For a prime $p \nmid 2$ let $d = \mathrm{ord}_p(2)$. Then $p \mid 2^d - 1$. Thus the first support condition would follow from the divisibility sequence implication

$$2^n - 1 \mid M^n - 1 \qquad (n \geq 1). \tag{S2}$$

It is enough that (S2) hold for every sufficiently large n: for fixed p, choose two consecutive integers $k, k+1$ with both $kd, (k+1)d$ in the valid range. Then $M^{kd} \equiv M^{(k+1)d} \equiv 1 \pmod p$, hence $M^d \equiv 1 \pmod p$.

Condition (S2) is far stronger than the original real equation, but it identifies a concrete algebraic bridge. It can be phrased on the 3-smooth monoid.

Definition 3.4 (Candidate monoid map)**.** Let

$$\mathcal{U} = \{2^a 3^b : a, b \in \mathbb{Z}_{\geq 0}\}.$$

Define the monoid homomorphism

$$\Phi : \mathcal{U} \longrightarrow \mathbb{Z}_{>0}, \qquad \Phi(2^a 3^b) = M^a A^b.$$

Definition 3.5 (Congruence preservation)**.** A map $f : E \to \mathbb{Z}$ on a subset $E \subset \mathbb{Z}$ is *congruence-preserving* if

$$u - v \mid f(u) - f(v) \qquad (u, v \in E).$$

Proposition 3.6 (Congruence-preservation criterion)**.** *If the candidate map Φ is congruence-preserving on $\mathcal{U}$, then $x \in \mathbb{Z}$. The same conclusion holds if congruence preservation is known only for the pairs $(2^n, 1)$ for all sufficiently large n.*

Proof. Since $1, 2^n \in \mathcal{U}$ and $\Phi(1) = 1$, congruence preservation gives

$$2^n - 1 \mid \Phi(2^n) - \Phi(1) = M^n - 1.$$

Now apply theorem 3.3, with the preceding two-consecutive-multiples argument in the eventual case. □

Exact new target. It is unnecessary to attack the four-exponentials statement in full generality. It would suffice to prove the single divisibility law $u - v \mid \Phi(u) - \Phi(v)$ on the sparse monoid of 3-smooth integers, or even only the one-dimensional family $2^n - 1 \mid M^n - 1$. Sections 7–9 show both why ordinary additive identities do not yield this law and how p-orderings could encode the missing divisibility.

3.2 Positive-density support failures from Kummer theory

The support theorem gives a qualitative dichotomy. For rational integers one can also see the failure set directly and compute its density.

Theorem 3.7 (Pairwise Kummer discordance)**.** *Let $u, v \in \mathbb{Q}^\times_{>0}$ be multiplicatively independent. For all but finitely many odd primes q, among rational primes $p \equiv 1 \pmod q$ the four events*

event	*conditional density among $p \equiv 1 \pmod q$*
u and v are qth powers modulo p	$1/q^2$
u is a qth power, v is not	$(q-1)/q^2$
v is a qth power, u is not	$(q-1)/q^2$
neither is a qth power	$(q-1)^2/q^2$

have the indicated Dirichlet densities. Their unconditional densities are respectively

$$\frac{1}{(q-1)q^2}, \qquad \frac{1}{q^2}, \qquad \frac{1}{q^2}, \qquad \frac{q-1}{q^2}.$$

Proof. For an admissible q, the classes of u and v are independent in $\mathbb{Q}(\zeta_q)^\times/(\mathbb{Q}(\zeta_q)^\times)^q$, so the Kummer extension

$$\mathbb{Q}(\zeta_q, u^{1/q}, v^{1/q})/\mathbb{Q}(\zeta_q)$$

has Galois group $\mathbb{F}_q^2$. For $p \equiv 1 \pmod q$, the two qth-power residue characters are the two coordinates of Frobenius. Chebotarev makes this vector uniform in $\mathbb{F}_q^2$. A coordinate is zero exactly when the corresponding rational number is a qth power modulo p. Counting zero/nonzero coordinates gives the conditional densities. Multiplication by the density $1/(q-1)$ of $p \equiv 1 \pmod q$ gives the unconditional values. □

Take $u = 2$ and $v = M$. If M is not a power of 2, the pair is multiplicatively independent. For every admissible q, theorem 3.7 supplies a positive-density set of primes such that 2 is a qth power but M is not. At each such prime, with $n = (p-1)/q$,

$$2^n \equiv 1 \pmod p, \qquad M^n \not\equiv 1 \pmod p.$$

Thus every admissible q produces explicit failures of $2 \preccurlyeq M$. The reverse event also has positive density, so the local orders of 2 and M are generically incomparable.

Corollary 3.8 (Rational support theorem, density form)**.** *For $M \in \mathbb{Z}_{>0}$, the following are equivalent:*

(i) *M is a nonnegative integral power of 2;*

(ii) $2 \preccurlyeq M$*;*

(iii) *for one, equivalently every, sufficiently large admissible prime q, the event "2 is a qth power modulo p but M is not" has density zero among $p \equiv 1 \pmod q$.*

This is a special rational case of the support theorem with additional quantitative information. Its limitation is equally clear: the real identity $M = 2^x$ gives no known reason for the qth-power status of M to follow that of 2 when x is not integral. The next section measures the stronger requirement that the two output residues arise from one common exponent.

4 Prime-power residue logarithms and common-exponent coherence

4.1 Residue-log vectors

Fix an admissible odd prime q, an integer $e \geq 1$, and put

$$Q = q^e, \qquad R_Q = \mathbb{Z}/Q\mathbb{Z}.$$

Let $p \equiv 1 \pmod Q$ be a prime not dividing $6MA$. Choose an element $\omega_p \in \mathbb{F}_p^\times$ of exact order Q. For $g \in \{2, 3, M, A\}$ define $\lambda_{p,Q}(g) \in R_Q$ by

$$g^{(p-1)/Q} = \omega_p^{\lambda_{p,Q}(g)}. \tag{4.1}$$

Replacing ω_p by ω_p^u, with $u \in R_Q^\times$, multiplies all four coordinates by u^{-1}. Consequently any condition invariant under simultaneous multiplication by a unit is intrinsic.

Write

$$\boldsymbol{\lambda}_{p,Q} = (\lambda_{p,Q}(2), \lambda_{p,Q}(3), \lambda_{p,Q}(M), \lambda_{p,Q}(A)).$$

Kummer reciprocity identifies this vector, up to the harmless common unit, with the Frobenius of a prime of K_Q in L_Q/K_Q.

Lemma 4.1 (Uniform residue-log distribution)**.** *Let $E \subseteq \mathrm{Gal}(L_Q/K_Q)$ be invariant under multiplication of all coordinates by units in $R_Q^\times$. Then the set of rational primes $p \equiv 1 \pmod Q$ for which $\boldsymbol{\lambda}_{p,Q} \in E$ has conditional Dirichlet density*

$$\frac{|E|}{|\,\mathrm{Gal}(L_Q/K_Q)|}.$$

Proof. Chebotarev makes Frobenius elements of prime ideals of K_Q uniform in the abelian group $\mathrm{Gal}(L_Q/K_Q)$. A rational prime $p \equiv 1 \pmod{Q}$ splits completely in K_Q. Its $\varphi(Q)$ primes correspond to the choices of the image of ζ_Q, hence to replacing ω_p by ω_p^u for $u \in R_Q^\times$. Since E is invariant under this common rescaling, either all primes of K_Q above p satisfy the condition or none do. The prime-ideal Chebotarev density therefore descends unchanged to the conditional density of rational primes. □

4.2 The coherence condition

Definition 4.2 (Q-coherence)**.** A prime $p \equiv 1 \pmod{Q}$ is *Q-coherent* for the quadruple $(2, 3, M, A)$ if there exists $k \in R_Q$ such that

$$\begin{aligned} \lambda_{p,Q}(M) &= k\lambda_{p,Q}(2), \\ \lambda_{p,Q}(A) &= k\lambda_{p,Q}(3). \end{aligned} \tag{4.2}$$

This condition is invariant under the choice of ω_p.

If there is an actual integer k satisfying $M \equiv 2^k \pmod{p}$ and $A \equiv 3^k \pmod{p}$, then (4.2) follows after raising to $(p-1)/Q$. The converse need not hold: Q-coherence sees only a quotient of the full discrete logarithm group. It is therefore a deliberately coarse but tractable necessary condition for a common finite-field exponent.

If $x = k \in \mathbb{Z}$, the equalities $M = 2^k$, $A = 3^k$ hold over $\mathbb{Z}$, so every prime $p \nmid 6MA$ is Q-coherent for every $Q \mid p-1$. The nonintegral branches behave in the opposite way.

4.3 Exact count in multiplicative rank four

Theorem 4.3 (Rank-four coherence density)**.** *Assume* $\operatorname{rank}\Gamma = 4$. *Let q be admissible and $Q = q^e$. Define*

$$N_4(q,e) = 1 + \frac{q(q^2-1)(q^{3e}-1)}{q^3-1}. \tag{4.3}$$

Then the conditional Dirichlet density of Q-coherent primes among $p \equiv 1 \pmod{Q}$ is

$$P_4(q,e) = \frac{N_4(q,e)}{q^{4e}}. \tag{4.4}$$

The unconditional density is

$$\frac{N_4(q,e)}{\varphi(Q)q^{4e}}.$$

Moreover,

$$P_4(q,e) \sim \frac{q(q+1)}{q^2+q+1}\, q^{-e} \qquad (e \to \infty). \tag{4.5}$$

Proof. By theorems 2.6 and 4.1, the residue-log vector is uniform in R_Q^4. Write its base and output halves as

$$\boldsymbol{b} = (u, v), \qquad \boldsymbol{y} = (m, a).$$

For fixed $\boldsymbol{b}$, the coherent outputs are exactly the cyclic submodule $R_Q\boldsymbol{b}$. If $\boldsymbol{b} = 0$, this submodule has one element. Otherwise define its content s by

$$s = \min\{v_q(u), v_q(v), e\}, \qquad 0 \le s < e.$$

Put $t = e - s$. There are

$$q^{2t} - q^{2t-2}$$

base vectors of content s, and the cyclic submodule they generate has q^t elements. Hence the number of coherent quadruples is

$$\begin{aligned} N_4(q,e) &= 1 + \sum_{t=1}^{e} (q^{2t} - q^{2t-2}) q^t \\ &= 1 + \frac{q(q^2-1)(q^{3e}-1)}{q^3-1}. \end{aligned}$$

Division by $|R_Q^4| = q^{4e}$ and application of theorem 4.1 gives the conditional density. The primes $p \equiv 1 \pmod{Q}$ have density $1/\varphi(Q)$, giving the unconditional value. The leading term in (4.3) yields (4.5). □

At depth one,

$$P_4(q,1) = \frac{q^3 - q + 1}{q^4} = \frac{1}{q} - \frac{1}{q^3} + \frac{1}{q^4}. \tag{4.6}$$

4.4 Exact count in multiplicative rank three

The rank-three relation cuts the ambient Kummer group by one linear equation, but the coherence variety also intersects that hyperplane nontransversely at imprimitive base vectors. The exact prime-power count therefore differs from simply multiplying the rank-four probability by q.

Theorem 4.4 (Rank-three coherence density)**.** *Assume* $\operatorname{rank}\Gamma = 3$*, let* $\boldsymbol{c}$ *be the primitive relation, and let* q *be admissible in the sense of theorem 2.5. Put* $Q = q^e$ *and*

$$N_3(q,e) = 1 + \frac{q(q^{2e}-1)}{q+1}. \tag{4.7}$$

Then the conditional Dirichlet density of Q*-coherent primes among* $p \equiv 1 \pmod{Q}$ *is*

$$P_3(q,e) = \frac{N_3(q,e)}{q^{3e}}. \tag{4.8}$$

The unconditional density is

$$\frac{N_3(q,e)}{\varphi(Q) q^{3e}},$$

and

$$P_3(q,e) \sim \frac{q}{q+1}\, q^{-e} \qquad (e \to \infty). \tag{4.9}$$

Proof. The Kummer group is the hyperplane

$$H_e = \{(u,v,m,a) \in R_Q^4 : c_1 u + c_2 v + c_3 m + c_4 a = 0\},$$

of cardinality q^{3e}. A coherent vector has the form (u, v, ku, kv). Thus it lies in H_e precisely when

$$(c_1 + c_3 k)u + (c_2 + c_4 k)v = 0 \pmod{Q}. \tag{4.10}$$

The zero base vector contributes one coherent vector.

Now let (u, v) have content $s < e$ and put $t = e - s$. Divide by q^s and work modulo q^t with a primitive vector (u', v'). Set

$$\alpha = c_1 u' + c_2 v', \qquad \delta = c_3 u' + c_4 v'.$$

Because $q \nmid \Delta(\boldsymbol{c})$, the matrix

$$\begin{pmatrix} c_1 & c_2 \\ c_3 & c_4 \end{pmatrix}$$

is invertible modulo q^t. Therefore $(u', v') \mapsto (\alpha, \delta)$ permutes the primitive vectors of $(\mathbb{Z}/q^t\mathbb{Z})^2$. Equation (4.10) becomes

$$\alpha + k\delta = 0 \pmod{q^t}.$$

For a primitive pair (α, δ) this has a solution exactly when δ is a unit, and then the solution k modulo q^t is unique. The number of such pairs is

$$q^t(q^t - q^{t-1}) = q^{2t} - q^{2t-1}.$$

Each gives one distinct coherent output. Summing over contents,

$$\begin{aligned} N_3(q, e) &= 1 + \sum_{t=1}^{e} (q^{2t} - q^{2t-1}) \\ &= 1 + \frac{q(q^{2e} - 1)}{q + 1}. \end{aligned}$$

Divide by $|H_e| = q^{3e}$ and invoke theorem 4.1. The remaining assertions follow as in the rank-four case. □

At depth one,

$$P_3(q, 1) = \frac{q^2 - q + 1}{q^3} = \frac{1}{q} - \frac{1}{q^2} + \frac{1}{q^3}. \tag{4.11}$$

For every $q > 1$, $P_4(q, 1) - P_3(q, 1) = (q - 1)^2/q^4 > 0$; the rank-four branch is slightly more coherent because there is no relation hyperplane to constrain the output.

4.5 The density gap

Theorem 4.5 (Finite-place density gap)**.** *Let (M, A) arise from a candidate exponent.*

(a) If $x \in \mathbb{Z}$, then for every Q and every prime $p \equiv 1 \pmod{Q}$ with $p \nmid 6MA$, the prime p is Q-coherent.

(b) If $x \notin \mathbb{Z}$, then $\operatorname{rank} \Gamma \in \{3, 4\}$, and for every admissible q and every $e \geq 1$ the conditional density of q^e-coherent primes is exactly $P_3(q, e)$ or $P_4(q, e)$ according to the rank branch. In particular it is $O_q(q^{-e})$, while the incoherent density tends to 1 as $e \to \infty$.

Proof. Part (a) follows by taking the common scalar $k = x \bmod Q$ in (4.2). Part (b) combines theorems 2.3, 4.3 and 4.4. □

Corollary 4.6 (Quantitative sufficient criterion)**.** *Fix an admissible q and $e \geq 1$. If one can prove, from the candidate equations, that the conditional density of q^e-coherent primes is strictly larger than both $P_3(q, e)$ and $P_4(q, e)$, then $x \in \mathbb{Z}$.*

The criterion is deliberately quantitative. A statement such as "there are infinitely many coherent primes" is useless: both nonintegral rank branches have positive density at every fixed depth. One needs a density excess, coherence at almost every admissible prime, or a mechanism that remains coherent through unbounded depth.

Interpretation. A hypothetical counterexample would satisfy the exact real determinant $\log 2 \, \log A - \log 3 \, \log M = 0$, yet at depth q^e its corresponding finite residue vectors would fail to share a scalar at a proportion tending to 1. The conjecture asks for a bridge showing that this extreme archimedean/finite-place mismatch cannot occur for the special integer quadruple. Chebotarev quantifies the mismatch perfectly; it does not remove it.

5 Recovering the rank branch from finite residue patterns

The depth-one Kummer data contains more than the aggregate coherence probability. In rank four, the four qth-power statuses are independent. In rank three, the unique multiplicative relation forbids entire status patterns. This gives a finite-place detector for the relation support.

Fix an admissible prime q and a prime $p \equiv 1 \pmod q$. A coordinate $\lambda_{p,q}(g)$ is zero exactly when g is a qth power modulo p. Let

$$J_q = \{i \in \{1,2,3,4\} : c_i \not\equiv 0 \pmod q\}$$

be the active support of the rank-three relation modulo q.

Lemma 5.1 (Nonzero solutions of one linear equation)**.** *For $s \geq 0$, let $N_s(q)$ be the number of $(z_1, \ldots, z_s) \in (\mathbb{F}_q^\times)^s$ satisfying*

$$z_1 + \cdots + z_s = 0.$$

Then

$$N_s(q) = \frac{(q-1)^s + (q-1)(-1)^s}{q}. \tag{5.1}$$

In particular $N_0(q) = 1$ and $N_1(q) = 0$.

Proof. Using additive-character orthogonality,

$$N_s(q) = \frac{1}{q} \sum_{t \in \mathbb{F}_q} \left(\sum_{z \in \mathbb{F}_q^\times} \exp \frac{2\pi i \, \mathrm{Tr}(tz)}{q} \right)^s.$$

The inner sum is $q-1$ for $t = 0$ and -1 for each $t \neq 0$, yielding (5.1). Equivalently, one may prove the same formula by the recurrence $N_s = (q-1)^{s-1} - N_{s-1}$. □

Theorem 5.2 (Rank-three status-pattern law)**.** *Assume* $\operatorname{rank} \Gamma = 3$ *and let q be admissible. Prescribe, for each of the four generators, whether its residue-log coordinate is zero or nonzero. Let*

- *s be the number of active coordinates in J_q prescribed to be nonzero;*
- *h be the number of inactive coordinates outside J_q prescribed to be nonzero.*

Then the number of points in the Kummer hyperplane realizing the pattern is

$$N_s(q)(q-1)^h. \tag{5.2}$$

Its conditional density among primes $p \equiv 1 \pmod{q}$ *is*

$$\frac{N_s(q)(q-1)^h}{q^3}. \tag{5.3}$$

In particular, every pattern with exactly one active nonzero coordinate is forbidden.

Proof. The residue vector is uniform on the hyperplane $\boldsymbol{c} \cdot \boldsymbol{t} = 0$ by theorem 4.1. After independently scaling each active coordinate by the nonzero coefficient c_i, the relation is a sum of the s prescribed nonzero active variables. There are $N_s(q)$ choices by theorem 5.1. Inactive nonzero coordinates are unrestricted and contribute $(q-1)^h$. The hyperplane has q^3 points. □

Proposition 5.3 (Rank-four status law)**.** *If* $\operatorname{rank} \Gamma = 4$*, every zero/nonzero pattern occurs. A pattern with* z *zero coordinates has conditional density*

$$\frac{(q-1)^{4-z}}{q^4}.$$

Thus the number of zero coordinates is binomial with parameters 4 *and* $1/q$*.*

The forbidden-pattern law is stronger than a generic rank computation: by varying q, observed forbidden patterns reveal which coefficients c_i vanish modulo q, and the frequencies of the remaining patterns constrain the relation support. With full residue-log values rather than only zero/nonzero statuses, one can recover the projective relation $[\boldsymbol{c}] \in \mathbb{P}^3(\mathbb{F}_q)$ from the orthogonal hyperplane. Chinese remaindering over several admissible q then reconstructs $\boldsymbol{c}$ once a height bound is available.

Example 5.4 (A clean rank-three test tuple)**.** The quadruple $(2, 3, 7, 42)$ has relation

$$2 \cdot 3 \cdot 7 = 42,$$

so $\boldsymbol{c} = (1, 1, 1, -1)$ and every coordinate is active for $q = 5$. The four patterns with exactly one non-5th-power coordinate are therefore forbidden. In the computation of section 6, each occurs zero times among 87,062 primes $p \le 5{,}000{,}000$ with $p \equiv 1 \pmod{5}$.

6 Exact computational checks

All experiments in this section use exact integer modular arithmetic. They are included for three reasons: to catch count-normalization errors, to test the descent from prime ideals to rational primes, and to provide small local witnesses suitable for later formalization. They do not test whether a candidate counterexample exists.

6.1 Direct finite-ring enumeration

Before sampling primes, the two numerator formulas were checked by exhaustive enumeration of all quadruples over $\mathbb{Z}/q^e\mathbb{Z}$. In the rank-three column the hyperplane was $u + v + m - a = 0$, corresponding to the test relation $2 \cdot 3 \cdot 7 = 42$.

q	e	exhaustive N_4	formula $N_4(q,e)$	exhaustive/formula N_3
3	1	25	25	7/7
3	2	673	673	61/61
5	1	121	121	21/21
5	2	15,121	15,121	521/521

This check is independent of prime generation and directly tests the zero-divisor cases at $e = 2$.

6.2 Pairwise discordance

Take $(u, v) = (2, 14)$ and $q = 5$. Neither generator is divisible by 5, and their classes are independent. Among primes $p \le 5{,}000{,}000$ with $p \equiv 1 \pmod 5$, the observed counts are:

Status modulo p	Count	Frequency	Predicted frequency
both 5th powers	3,339	0.03835198	0.04000000
2 a 5th power, 14 not	14,006	0.16087386	0.16000000
14 a 5th power, 2 not	13,980	0.16057522	0.16000000
neither a 5th power	55,737	0.64019894	0.64000000
Total	87,062	1	1

For several values of q, the least prime witnesses found below the same limit are shown next. Here $n = (p-1)/q$; the first witness has $2^n \equiv 1$ and $14^n \not\equiv 1$, while the second has the reverse status.

q	first p	first n	reverse p	reverse n
3	31	10	13	4
5	151	30	41	8
7	631	90	71	10
11	331	30	397	36
13	4421	340	547	42
17	1429	84	2347	138
19	1103	58	1597	84

These are concrete finite certificates of the two support failures.

6.3 Common-exponent coherence at depths 5 and 25

Two toy quadruples isolate the two rank branches:

$$(2, 3, 7, 11) \quad \text{has rank } 4,$$

while

$$(2, 3, 7, 42) \quad \text{has rank } 3, \qquad 2 \cdot 3 \cdot 7 = 42.$$

The next table compares the observed coherence frequencies with theorems 4.3 and 4.4. The $Q = 5$ rows use primes at most 5,000,000; the $Q = 25$ rows use primes at most 10,000,000. The z column is the deviation divided by the binomial standard deviation $\sqrt{NP(1-P)}$.

Rank	Q	Eligible primes	Coherent	Observed	Predicted	z
4	5	87,061	16,760	0.19250870	0.19360000	−0.81
3	5	87,062	14,476	0.16627231	0.16800000	−1.36
4	25	33,155	1,281	0.03863671	0.03870976	−0.07
3	25	33,155	1,098	0.03311718	0.03334400	−0.23

The depth-25 predicted numerators are

$$N_4(5, 2) = 15{,}121, \qquad N_3(5, 2) = 521.$$

The agreement is especially useful because prime-power rings contain zero divisors; a mistaken field-only count would generally first appear at $e = 2$.

6.4 Status-pattern checks

For the rank-four tuple at $q = 5$, grouping by the number of coordinates that are 5th powers gives:

Number of 5th-power coordinates	Count	Observed	Predicted
0	35,627	0.40921882	0.40960000
1	35,800	0.41120594	0.40960000
2	13,354	0.15338671	0.15360000
3	2,143	0.02461493	0.02560000
4	137	0.00157361	0.00160000

For the rank-three tuple, all four one-nonresidue patterns

$$1110, \quad 1101, \quad 1011, \quad 0111$$

(where 1 means "5th power") had count zero, exactly as theorem 5.2 predicts. Selected nonzero patterns were:

Pattern	Count	Observed	Predicted
0000	36,185	0.41562335	0.41600000
0001	8,387	0.09633365	0.09600000
0011	2,781	0.03194275	0.03200000
1111	649	0.00745446	0.00800000

The maximum absolute frequency error over all sixteen patterns was 0.001103.

6.5 What the computations do and do not establish

The tests strongly validate the finite counts and implementation. They do not add an assumption such as GRH, because the theoretical densities are unconditional consequences of Chebotarev. They also do not provide statistical evidence for the Alaoglu–Erdős conjecture: the toy tuples were chosen solely to have known multiplicative rank. Their role is proof engineering.

7 The additive skeleton of the 3-smooth monoid

The support criterion in theorem 3.6 asks for congruence preservation on $\mathcal{U} = \{2^a 3^b\}$. A natural first attempt is to exploit additive identities in $\mathcal{U}$ and transport them through Φ. This section classifies all such identities and proves that the attempt collapses to one fixed modulus.

7.1 Classification of primitive additive triples

Call a solution $u + v = w$ in $\mathcal{U}$ *primitive* if $\gcd(u, v, w) = 1$.

Theorem 7.1 (Complete primitive additive skeleton)**.** *Up to interchanging u and v, the primitive solutions of*

$$u + v = w, \qquad u, v, w \in \mathcal{U},$$

are exactly

$$1 + 1 = 2, \qquad 1 + 2 = 3, \qquad 1 + 3 = 4, \qquad 1 + 8 = 9. \tag{7.1}$$

Every nonprimitive solution is obtained by multiplying one of these four by an element of $\mathcal{U}$.

Proof. In a primitive solution the three numbers are pairwise coprime, since $\gcd(u,w) = \gcd(u,v)$ and similarly for v. If both u and v are greater than one, their prime supports must be disjoint, so up to order $u = 2^a$, $v = 3^b$ with $a, b > 0$. But then w is coprime to both 2 and 3, forcing $w = 1$, impossible. Thus one summand is 1.

The case $1 + 1 = 2$ is immediate. The remaining equations are

$$3^b - 2^a = 1 \qquad \text{or} \qquad 2^a - 3^b = 1.$$

For the first, $a = 1$ gives $(a, b) = (1, 1)$. If $a \geq 2$, reduction modulo 4 makes b even, say $b = 2m$. Then

$$(3^m - 1)(3^m + 1) = 2^a.$$

Both factors are powers of 2, have gcd 2, and differ by 2; hence they are 2 and 4. Thus $m = 1$, giving $(a, b) = (3, 2)$.

For the second equation, reduction modulo 3 makes a even, say $a = 2m$. Then

$$(2^m - 1)(2^m + 1) = 3^b.$$

The coprime factors are powers of 3 differing by 2, so they are 1 and 3. Hence $m = 1$ and $(a, b) = (2, 1)$. This proves (7.1). Dividing any nonprimitive solution by its common gcd, which is again 3-smooth, gives the final assertion. □

No use of Catalan–Mihăilescu is needed; the two special exponential equations factor after a single congruence observation.

7.2 Defects of the four primitive relations

Transport the four identities through Φ. With signs chosen for convenience, define

$$\begin{aligned} d_{11} &= M - 2, \\ d_{12} &= A - M - 1, \\ d_{13} &= M^2 - A - 1, \\ d_{18} &= A^2 - M^3 - 1. \end{aligned} \tag{7.2}$$

They satisfy the exact identities

$$d_{13} = (M + 1)d_{11} - d_{12}, \tag{7.3}$$

and

$$d_{18} = -M(M + 1)d_{11} + \big(2(M + 1) + d_{12}\big)d_{12}. \tag{7.4}$$

Moreover

$$d_{12} = (A - 3) - (M - 2).$$

Thus all four primitive defects generate the same ideal as $M - 2$ and $A - 3$.

Theorem 7.2 (Additive defect ideal)**.** *Let $\mathcal{D}$ be the ideal of $\mathbb{Z}$ generated by all transported additive defects*

$$\Phi(u) + \Phi(v) - \Phi(w)$$

with $u + v = w$ in $\mathcal{U}$. Then

$$\mathcal{D} = (M - 2, A - 3) = C\mathbb{Z}, \qquad C := \gcd(M - 2, A - 3). \tag{7.5}$$

Proof. By theorem 7.1, every additive triple is a 3-smooth multiple of one of the four primitive triples. Under Φ, multiplication by 2^a3^b multiplies the defect by M^aA^b. Hence all defects lie in the ideal generated by the four numbers in (7.2). Equations (7.3)–(7.4) reduce that ideal to (d_{11}, d_{12}), which equals $(M-2, A-3)$. Conversely, d_{11} and d_{12} themselves come from the first two primitive identities, so equality holds. □

The next theorem shows that allowing arbitrary signed polynomial combinations of additive identities creates no new modulus.

Theorem 7.3 (Universal polynomial collapse)**.** *For every $P \in \mathbb{Z}[X, Y]$ with $P(2, 3) = 0$,*

$$C \mid P(M, A). \tag{7.6}$$

Conversely, the ideal generated by all integers $P(M, A)$ with $P(2, 3) = 0$ is exactly $C\mathbb{Z}$. Equivalently, C is the largest positive integer D for which

$$\Phi(u) \equiv u \pmod{D} \qquad \text{for every } u \in \mathcal{U}. \tag{7.7}$$

Proof. The evaluation kernel at $(2, 3)$ is

$$\ker\bigl(\mathbb{Z}[X, Y] \xrightarrow{P \mapsto P(2,3)} \mathbb{Z}\bigr) = (X-2, Y-3).$$

Thus $P = (X-2)U + (Y-3)V$ for some $U, V \in \mathbb{Z}[X, Y]$, and $P(M, A)$ is divisible by C. Conversely, $P = X-2$ and $P = Y-3$ show that the generated ideal contains $(M-2, A-3) = C\mathbb{Z}$.

Since $M \equiv 2$ and $A \equiv 3 \pmod{C}$, every monomial satisfies $M^aA^b \equiv 2^a3^b \pmod{C}$, proving (7.7) for $D = C$. Any modulus with that property must divide both $M-2$ and $A-3$, hence divide C. □

Audit warning 7.4. The conclusion is not that additive information is useless. It is that every identity obtained solely by evaluating an integral polynomial relation true at $(2, 3)$ can see only the fixed anchor C. A proof of congruence preservation must use varying differences $u - v$, multiplicative orders, or prime-dependent geometry; it cannot arise from accumulating more polynomial identities at the single point $(2, 3)$.

8 Dynamic anchors and a subexponential no-go theorem

The fixed modulus C of theorem 7.3 compares (M, A) with $(2, 3)$ at one time. A stronger attempt compares the nth powers and allows the modulus to vary:

$$C_n := \gcd(M^n - 2^n, A^n - 3^n). \tag{8.1}$$

For a nonintegral candidate neither difference is identically zero.

Proposition 8.1 (Dynamic polynomial collapse)**.** *For each $n \geq 1$, the ideal generated by*

$$\{P(M^n, A^n) : P \in \mathbb{Z}[X, Y],\ P(2^n, 3^n) = 0\}$$

is exactly $C_n\mathbb{Z}$. Equivalently, C_n is the largest modulus D for which

$$M^{an}A^{bn} \equiv 2^{an}3^{bn} \pmod{D} \qquad (a, b \geq 0).$$

Moreover, if $m \mid n$, then $C_m \mid C_n$.

Proof. Apply the proof of theorem 7.3 with the evaluation point $(2^n, 3^n)$. Its kernel is $(X-2^n, Y-3^n)$, and evaluation at (M^n, A^n) produces the ideal $(M^n-2^n, A^n-3^n) = C_n\mathbb{Z}$. If $m \mid n$, then $X^m - Y^m$ divides $X^n - Y^n$, so both generators of the mth ideal divide their nth counterparts. □

For a prime $p \nmid 6MA$,

$$p \mid C_n \iff \left(\frac{M}{2}\right)^n \equiv 1 \pmod p \text{ and } \left(\frac{A}{3}\right)^n \equiv 1 \pmod p. \tag{8.2}$$

Thus C_n records simultaneous return times of the two rational ratios $M/2$ and $A/3$.

Lemma 8.2 (Independence of the dynamic ratios)**.** *If $x \neq 1$, the rational numbers $M/2$ and $A/3$ are multiplicatively independent.*

Proof. A relation $(M/2)^r(A/3)^s = 1$ gives, after substituting $M = 2^x$, $A = 3^x$ and taking real logarithms,

$$(x-1)(r\log 2 + s\log 3) = 0.$$

Since $x \neq 1$ and $2, 3$ are multiplicatively independent, $r = s = 0$. □

Theorem 8.3 (Subexponential dynamic anchors)**.** *For a nonintegral candidate,*

$$\log C_n = o(n) \qquad (n \to \infty). \tag{8.3}$$

Equivalently, for every $\varepsilon > 0$, one has $C_n < e^{\varepsilon n}$ for all sufficiently large n.

Proof. Put $U = M/2$ and $V = A/3$. By theorem 8.2, U and V are multiplicatively independent rational S-units for the fixed finite set of primes dividing $6MA$. The Corvaja–Zannier S-unit gcd estimate, extending the Bugeaud–Corvaja–Zannier theorem, gives an $o(n)$ upper bound for the contribution outside S to the generalized gcd of $U^n - 1$ and $V^n - 1$.

It remains to control primes in S. If a prime $p \notin \{2,3\}$ divides M, then $v_p(M^n - 2^n) = 0$; if it divides A, then $v_p(A^n - 3^n) = 0$. Thus such primes contribute nothing to the common gcd unless both relevant terms are p-adic units, in which case the standard lifting-the-exponent argument gives $O(v_p(n)+1) = O(\log n)$. At $p = 2$, either A is even and $A^n - 3^n$ is odd, or A is odd and the same lifting argument gives $v_2(A^n - 3^n) = O(\log n)$. The prime 3 is symmetric, using $M^n - 2^n$. Hence the total contribution from the fixed set S is $O(\log n)$. Combining it with the outside-S estimate proves (8.3). □

Remark 8.4. The theorem also holds for integral exponents $x = k > 1$; it is a limitation theorem for the anchor method, not an integer/noninteger discriminator. The exceptional exponent $x = 1$ makes both differences zero.

8.1 Why the additive route stops

A support proof along (S2) would have to transfer all prime divisors of 2^n-1 to M^n-1. The modulus $2^n - 1$ has exponential logarithmic size. By contrast, every modulus produced by simultaneous polynomial return of (M^n, A^n) to $(2^n, 3^n)$ divides C_n, whose logarithm is $o(n)$. Thus there is an exponential gap:

$$\log(2^n - 1) \sim n\log 2, \qquad \log C_n = o(n).$$

This does not refute all uses of addition. It refutes the specific strategy of accumulating ordinary polynomial identities at the two evaluation points and hoping their common defect becomes large enough to enforce support. Any successful additive argument must retain prime-by-prime order information that the single gcd C_n discards.

9 Local congruence kernels and the Bhargava-ordering route

This section converts global congruence preservation into exact finite cyclic-group statements. It then explains how p-orderings offer a disciplined way to attack those statements without claiming that the existing theory solves them automatically.

9.1 Kernel inclusion on exponent lattices

For an odd prime $p \nmid 6MA$ and $m \geq 1$, define homomorphisms

$$\begin{aligned} \rho_{p^m} &: \mathbb{Z}^2 \longrightarrow (\mathbb{Z}/p^m\mathbb{Z})^\times, \quad (a,b) \longmapsto 2^a 3^b, \\ \rho'_{p^m} &: \mathbb{Z}^2 \longrightarrow (\mathbb{Z}/p^m\mathbb{Z})^\times, \quad (a,b) \longmapsto M^a A^b. \end{aligned} \tag{9.1}$$

Although Φ was defined only for nonnegative exponents, any difference in $\mathbb{Z}^2$ can be realized by adding a sufficiently large common vector to two nonnegative exponent pairs. Consequently the local congruence condition on $\mathcal{U}$ is exactly a kernel inclusion.

Proposition 9.1 (Local kernel criterion)**.** *For fixed $p \nmid 6MA$ and $m \geq 1$, the following are equivalent.*

(i) Whenever $u, v \in \mathcal{U}$ satisfy $u \equiv v \pmod{p^m}$, one has $\Phi(u) \equiv \Phi(v) \pmod{p^m}$.

(ii) $\ker \rho_{p^m} \subseteq \ker \rho'_{p^m}$.

(iii) The rule

$$2^a 3^b \longmapsto M^a A^b$$

defines a well-defined group homomorphism

$$f_{p^m} : \langle 2, 3 \rangle \longrightarrow (\mathbb{Z}/p^m\mathbb{Z})^\times.$$

Proof. The equivalence of (ii) and (iii) is the universal property of a quotient. To compare with (i), write $u = 2^a 3^b$ and $v = 2^c 3^d$. Since all four generators are units modulo p^m, $u \equiv v$ is equivalent to $(a-c, b-d) \in \ker \rho_{p^m}$, and the corresponding output congruence is membership in $\ker \rho'_{p^m}$. Negative coordinate differences cause no issue, as both exponent pairs may be shifted into $\mathbb{Z}^2_{\geq 0}$. ☐

For odd p, the unit group $(\mathbb{Z}/p^m\mathbb{Z})^\times$ is cyclic. This makes every local monoid homomorphism a common power map.

Theorem 9.2 (Local common-exponent criterion)**.** *Let p be an odd prime with $p \nmid 6MA$, and let $m \geq 1$. The equivalent conditions of theorem 9.1 hold if and only if there exists an integer $k_{p,m}$ such that*

$$M \equiv 2^{k_{p,m}} \pmod{p^m}, \qquad A \equiv 3^{k_{p,m}} \pmod{p^m}. \tag{9.2}$$

Proof. If (9.2) holds, raising congruent elements of $\mathcal{U}$ to the common power $k_{p,m}$ proves local congruence preservation.

Conversely, let $H = \langle 2, 3 \rangle$ in the cyclic group $G = (\mathbb{Z}/p^m\mathbb{Z})^\times$, and let $f : H \to G$ be the homomorphism from theorem 9.1. The image $f(H)$ has order dividing $|H|$. A cyclic group has a unique subgroup of each order dividing $|G|$, so every element whose order divides $|H|$ lies in H; hence $f(H) \subseteq H$. Thus f is an endomorphism of the cyclic group H, and every such endomorphism is $h \mapsto h^k$ for some integer k. Evaluating at 2 and 3 gives (9.2). ☐

Corollary 9.3 (Global congruence preservation as local exponents)**.** *Away from the finite set of primes dividing $6MA$, the map Φ is congruence-preserving if and only if compatible common exponents $k_{p,m}$ as in (9.2) exist for every p and m. For the support conclusion alone, it is enough to have a common exponent modulo p for almost all p.*

The depth-Q coherence of theorem 4.2 is a quotient shadow of (9.2): a full common exponent modulo p implies Q-coherence for every $Q \mid p-1$. The exact densities therefore quantify how rarely the first necessary quotient condition holds in the nonintegral rank branches.

9.2 Bhargava p-orderings as a concrete coordinate system

Bhargava introduced p-orderings of an arbitrary subset of a Dedekind domain to describe integer-valued polynomial functions and generalized factorials. For a subset $E \subset \mathbb{Z}$ and a prime p, a p-ordering is a sequence $e_0, e_1, \ldots$ chosen recursively so that e_n minimizes

$$v_p\left(\prod_{j<n}(e-e_j)\right)$$

among $e \in E$. The resulting valuation sequence is independent of the choices and defines the generalized factorial of E.

For $E = \mathcal{U}$, the residues are generated multiplicatively by 2 and 3. Modulo p^m, the geometry of E is therefore controlled by the lattice $\ker \rho_{p^m} \subset \mathbb{Z}^2$. This makes the ordering problem unusually explicit: a greedy choice is a problem of selecting exponent vectors whose images spread through the cyclic subgroup $\langle 2, 3\rangle$ with maximal p-adic separation.

The route proposed here has four precise stages.

1. **Compute the generalized factorials of $\mathcal{U}$.** Express their p-adic valuations in terms of the orders of 2 and 3 modulo p^m and the index of $\langle 2, 3\rangle$ in $(\mathbb{Z}/p^m\mathbb{Z})^\times$.
2. **Interpolate Φ on a p-ordering.** The Newton coefficients are divided differences of values M^aA^b. Their required divisibility can be stated entirely in $\mathbb{Z}_p$.
3. **Convert coefficient divisibility to kernel inclusion.** One seeks a theorem adapted to this sparse multiplicative set, showing that the special exponential form of the values forces the local condition in theorem 9.1.
4. **Use the density obstruction.** If that condition holds at almost all primes, theorem 9.2 gives full local common exponents, while theorem 4.5 rules out both nonintegral rank branches.

Audit warning 9.4. Bhargava's theory characterizes polynomial functions and integer-valued interpolation under specific hypotheses; it does not state that an arbitrary monoid homomorphism is congruence-preserving. The proposed use is a coordinate system and divisibility framework, not a citation that closes the problem.

Research target 9.5 (Sparse-monoid factorial problem)**.** Determine, for every odd prime p, the Bhargava factorial sequence of $\mathcal{U} = \{2^a3^b : a, b \geq 0\}$ in terms of the filtration

$$\ker \rho_p \supseteq \ker \rho_{p^2} \supseteq \cdots .$$

Then characterize those pairs (M, A) for which the map $2^a3^b \mapsto M^aA^b$ has Newton coefficients satisfying the congruence-preserving divisibility bounds.

This target is finite at every level, branch-sensitive through the Kummer relation, and directly connected to the exact support criterion. It is therefore better focused than a generic appeal to integer-valued polynomial theory.

10 Finite regulators, horizontal sieves, and the missing bridge

The exact density theorem solves the local distribution problem conditional only on multiplicative rank. What remains is to make the real identity communicate with those local characters. This section formulates that gap in three related ways.

10.1 An archimedean tensor killed by one bilinear functional

In the tensor square of the rational multiplicative group, consider

$$\tau = 2 \otimes A - 3 \otimes M \in \Gamma \otimes_{\mathbb{Z}} \Gamma. \tag{10.1}$$

The archimedean logarithm gives a bilinear functional

$$B_\infty(u \otimes v) = \log u \, \log v.$$

The candidate equations say exactly

$$B_\infty(\tau) = \log 2 \, \log A - \log 3 \, \log M = 0. \tag{10.2}$$

At a prime $p \equiv 1 \pmod{Q}$, the residue character $\lambda_{p,Q} : \Gamma \to R_Q$ gives the finite determinant

$$D_{p,Q} = \lambda_{p,Q}(2)\lambda_{p,Q}(A) - \lambda_{p,Q}(3)\lambda_{p,Q}(M). \tag{10.3}$$

This is the image of τ under the corresponding finite bilinear functional. Q-coherence implies $D_{p,Q} = 0$. At depth one, if the base vector $(\lambda(2), \lambda(3))$ is nonzero, the converse also holds because two vectors over a field are proportional exactly when their determinant vanishes. Over R_Q with $e > 1$, determinant vanishing is weaker because of zero divisors, which is why the cyclic-submodule count in theorems 4.3 and 4.4 is the correct object.

The conjectural missing principle can now be stated without mentioning exponentials.

Research target 10.1 (Finite-regulator principle)**.** Find an arithmetic reciprocity law, product formula, height inequality, or regulator construction for tensors such as (10.1) with the following consequence: if a tensor built from positive rational integers is killed by B_∞ for the special reason $\log M/\log 2 = \log A/\log 3$, then its finite Kummer images (10.3) vanish, or become coherent in the stronger cyclic-submodule sense, at a set of primes whose density exceeds the exact rank-three and rank-four baselines.

Ordinary product formulas apply to sums of valuations of one algebraic number; here the archimedean expression is a product of logarithms and the finite data consists of power-residue characters. A solution may therefore require a genuinely higher object—for example a cup product, a K_2 symbol, or an adelic regulator—rather than another linear form in logarithms. The point of theorem 10.1 is not to guess the correct cohomology in advance, but to give it a numerical acceptance test: it must beat $P_3(q, e)$ and $P_4(q, e)$.

10.2 Depth amplification

Because both coherence probabilities tend to zero, even a weak lower bound uniform in depth would settle the problem.

Corollary 10.2 (Uniform-depth amplification criterion)**.** *Suppose there exist an admissible prime q, a constant $\delta > 0$, and an unbounded set of integers e such that the candidate equations imply*

$$\underline{\mathrm{dens}}\left(\{p \equiv 1 \pmod{q^e} : p \text{ is } q^e\text{-coherent}\} \mid p \equiv 1 \pmod{q^e}\right) \geq \delta. \tag{10.4}$$

Then $x \in \mathbb{Z}$.

Proof. For a nonintegral candidate the exact conditional density is $P_3(q,e)$ or $P_4(q,e)$, both of which tend to zero. This contradicts (10.4) along the unbounded set of depths. □

This criterion is substantially weaker than congruence preservation. It does not ask for local common exponents at almost all primes, or even for density one. Any positive amount of coherence that survives arbitrarily deep q-power quotients is enough.

Possible sources of depth amplification include:

1. **A q-adic interpolation of the exponent.** Construct, for many primes p, a compatible scalar in $\mathbb{Z}_q$ from the two equations and show its reductions solve (4.2). The obstacle is that x is a real number with no given q-adic incarnation.
2. **A reciprocity constraint on the tensor τ.** Show that $B_\infty(\tau) = 0$ forces small finite regulator at all depths for a positive proportion of Frobenius elements.
3. **A congruence-preserving interpolation theorem.** Use the special values of Φ on a p-ordering of $\mathcal{U}$ to obtain local common exponents modulo p^m for a set of primes with uniform lower density.
4. **An unlikely-intersection theorem in character space.** The Kummer characters of Γ form R_Q^r; coherence is a union of cyclic submodules. One needs an arithmetic reason for the Frobenius orbit of the special candidate to hit this thin set more often than a generic rank-r point.

10.3 Horizontal Frobenius sieve

A full common exponent modulo p implies q-coherence for every prime $q \mid p-1$. Thus one may impose many independent-looking Kummer filters horizontally, across distinct q. For a single admissible q, the filter rejects a proportion

$$\frac{1}{q-1}\left(1 - P_r(q,1)\right)$$

of all primes at the first q-level. The sum of the leading rejection probabilities over q diverges like $\sum_q 1/q$. This suggests that a large-sieve theorem for the compatible Kummer representations should force the set

$$\mathcal{P}_{\mathrm{full}} = \{p : \exists k_p, M \equiv 2^{k_p},\ A \equiv 3^{k_p} \pmod{p}\} \tag{10.5}$$

to have density zero whenever $r > 2$.

A zero-density theorem for (10.5) would strengthen the qualitative support theorem and would be valuable independently of the conjecture. It still would not prove Alaoglu–Erdős, because no current argument places almost all primes in $\mathcal{P}_{\mathrm{full}}$. Its value is strategic: it converts any future positive-density local-exponent consequence of the real equations into an immediate contradiction.

The technical issue is entanglement. Kummer fields for different q share cyclotomic and occasionally radical subfields, so one cannot multiply the one-prime probabilities naively. The explicit entanglement-group theory of Perucca–Sgobba–Tronto is the natural tool for making the horizontal sieve rigorous.

Research target 10.3 (Horizontal common-exponent sieve)**.** Prove that if $\operatorname{rank}\langle 2,3,M,A\rangle > 2$, then the prime set $\mathcal{P}_{\text{full}}$ in (10.5) has natural density zero, with an effective upper-bound sieve expressed through the Kummer entanglement invariants of Γ.

10.4 Relation recovery as a computational attack on rank three

The rank-three branch is more structured and may be attacked separately. For admissible q, exact residue logs at sufficiently many primes identify the hyperplane $\boldsymbol{c}^{\perp} \subset \mathbb{F}_q^4$. Repeating this for several q reconstructs the primitive relation $\boldsymbol{c}$, after which the real exponent has the fractional-linear expression

$$x = -\frac{c_1 + c_2\theta}{c_3 + c_4\theta}, \qquad \theta = \frac{\log 3}{\log 2}. \tag{10.6}$$

The integrality conditions become the pair

$$2^{-\frac{c_1+c_2\theta}{c_3+c_4\theta}} \in \mathbb{Z}, \qquad 3^{-\frac{c_1+c_2\theta}{c_3+c_4\theta}} \in \mathbb{Z}.$$

This does not remove the four-exponentials wall, but it reduces the branch to a countable family indexed by primitive integer matrices with nonzero determinant. A branch-specific computation can therefore prioritize relations of small height, test all local pattern consequences, and search for additional congruences forced by the valuation supports of M and A.

A useful formal target is an effective lower bound for the height of $\boldsymbol{c}$ in terms of M, A, or a theorem excluding all relations with $|\Delta|$ in a prescribed range. Such a theorem would combine well with exact relation recovery modulo many q.

11 Synthesis: what would now constitute a proof

The new local theory organizes several sufficient routes by strength.

Route	Required statement	Why it concludes
Full congruence preservation	$u - v \mid \Phi(u) - \Phi(v)$ for all $u, v \in \mathcal{U}$	theorem 3.6 and the support theorem
One-dimensional divisibility	$2^n - 1 \mid M^n - 1$ for all sufficiently large n	Order support for $2, M$
Almost-everywhere local exponent	For almost all p, one k_p satisfies $M \equiv 2^{k_p}$ and $A \equiv 3^{k_p} \pmod p$	Implies $2 \preccurlyeq M$ and $3 \preccurlyeq A$
Fixed-depth density excess	For some admissible (q, e), coherence density exceeds both exact baselines	theorem 4.6
Uniform depth	A positive lower coherence density survives unbounded e	theorem 10.2
Finite regulator	$B_\infty(\tau) = 0$ forces enough finite determinant or cyclic-submodule vanishing	Reduces to either density criterion

The additive results supply a complementary exclusion table.

Input used	Maximum modulus obtained	Obstruction
All primitive equations $u+v=w$ in $\mathcal{U}$	$C=\gcd(M-2,A-3)$	Fixed in n
All $P(2,3)=0$ with $P\in\mathbb{Z}[X,Y]$	The same C	Universal ideal collapse
All $P(2^n,3^n)=0$ at time n	$C_n=\gcd(M^n-2^n,A^n-3^n)$	$\log C_n=o(n)$
Support target	Needs to carry divisors of size about $e^{n\log 2}$	Exponential gap

The synthesis is therefore sharp. The local side is not vague: its generic distributions and thresholds are exact. The additive side is not merely unsuccessful: a full ideal theorem and an S-unit gcd estimate explain its ceiling. The remaining problem is one specific transfer: produce non-generic finite-place coherence from the special real logarithmic dependence.

12 Lean formalization blueprint

A useful formalization should separate elementary algebra, finite combinatorics, and deep imported number theory. The following module graph minimizes trust boundaries.

12.1 Kernel-checkable elementary layer

The following statements can be formalized in Lean with Mathlib and no analytic number theory.

1. **Candidate normalization.** Rational candidate implies integral; rank-two exclusion assuming the transcendence of $\log 3/\log 2$ as an external theorem.

2. **Relation algebra.** Primitive relation lattice in rank three and $\Delta\neq 0$ for a nonintegral candidate.

3. **Finite-ring content.** Cardinality of vectors of exact q-adic content in $(\mathbb{Z}/q^e\mathbb{Z})^2$ and size of the cyclic submodule they generate.

4. **Rank-four count.** The identity
$$1+\sum_{t=1}^{e}(q^{2t}-q^{2t-2})q^t=1+\frac{q(q^2-1)(q^{3e}-1)}{q^3-1}.$$

5. **Rank-three count.** Invertible change of variables under $q\nmid\Delta$ and
$$1+\sum_{t=1}^{e}(q^{2t}-q^{2t-1})=1+\frac{q(q^{2e}-1)}{q+1}.$$

6. **Pattern law.** The recurrence and closed formula for $N_s(q)$, including the forbidden one-nonzero patterns.

7. **Additive skeleton.** The elementary factorization proof of the four primitive triples in (7.1).

8. **Defect identities and ideals.** Equations (7.3)–(7.4), the kernel $(X-2,Y-3)$, and the universal polynomial collapse.

9. **Dynamic divisibility.** $C_m \mid C_n$ when $m \mid n$.
10. **Local cyclic groups.** Kernel inclusion and the common-exponent criterion for cyclic unit groups modulo odd prime powers.

12.2 External theorem interfaces

The deep inputs should appear as explicit assumptions or imported theorems with narrow signatures.

Kummer realization
: For admissible q, identify the Galois group with the quotient of the exponent lattice computed in theorem 2.6.

Chebotarev
: For a finite abelian extension, assign density $|E|/|G|$ to a union of Frobenius elements, then prove the scalar-invariant descent from prime ideals of K_Q to rational primes.

Support theorem
: Use the Corrales–Rodrigáñez–Schoof implication in the one-dimensional torus.

S-unit gcd
: Package only the asymptotic statement needed for theorem 8.3; handle the finite support primes by formalized LTE lemmas.

This separation would already allow a machine-checked conditional theorem: *assuming the four external interfaces, the exact density gap and additive no-go results follow.* Full formalization of Chebotarev or the Subspace Theorem is not a prerequisite for checking the novel finite combinatorics.

12.3 Computational certificates

For fixed (q, e), the theoretical numerator identities can be certified by exhaustive enumeration in `ZMod`(q^e) for small parameters. Prime experiments should be represented as independent certificates containing:

- the prime list or a deterministic primality certificate;
- a primitive Qth root in each finite field;
- the four residue logs;
- the coherence witness k when it exists;
- the relation check in rank three.

The proof does not depend on these certificates, but they are excellent regression tests for a formal development.

13 Prioritized research program

The following order maximizes the chance of converting the new framework into a proof rather than another collection of analogies.

1. **Formalize the two coherence counts.** They are short, exact, and likely to reveal any hidden issue with the prime-power content stratification.

2. **Prove the horizontal zero-density theorem** in theorem 10.3. This is a natural standalone theorem about reductions of two points on $\mathbb{G}_m^2$ and can use existing entanglement machinery.
3. **Compute p-orderings of the 3-smooth monoid.** Begin with primes for which $\langle 2, 3\rangle$ has small index modulo p^m; search for a stable generalized-factorial formula.
4. **Test candidate-value divided differences symbolically.** Express the first several Newton coefficients as Laurent polynomials in M, A and factor their fixed divisors. The additive collapse theorem predicts which factors are spurious and which could encode orders.
5. **Develop a finite regulator.** Search first for a depth-one reciprocity law for $D_{p,q}$, where the determinant criterion is linear algebra over a field. Only after a genuine density excess appears should the construction be lifted to q^e.
6. **Exploit the rank-three pattern detector.** Combine relation reconstruction with valuation support to exclude low-height primitive relations; automate exact searches with proof-producing certificates.
7. **Maintain an adversarial claim ledger.** Any proposed bridge must explicitly state whether it yields infinitely many coherent primes, positive density, density one, or uniform depth. Only the last three can interact decisively with the exact baselines.

14 Conclusion

This investigation does not close the Alaoglu–Erdős conjecture, but it changes the shape of the remaining work.

First, the conjecture has an exact support-theoretic endpoint: it is enough to force $2 \preccurlyeq M$ or $3 \preccurlyeq A$. Congruence preservation of the candidate monoid map is a concrete sufficient mechanism, and locally it is equivalent to the existence of one common finite exponent for the pairs $(2, 3)$ and (M, A).

Second, the finite-place behavior of every hypothetical nonintegral candidate is now quantified. At prime-power depth q^e, common-exponent coherence has one of two exact densities, determined solely by whether the multiplicative rank is three or four. Both decay like q^{-e}, while an integral exponent has density one. The rank-three relation also leaves a visible fingerprint of forbidden qth-power patterns.

Third, the obvious additive route has been exhausted in a rigorous sense. The entire additive skeleton of the 3-smooth monoid consists of four primitive triples; all their defects, and in fact all integral polynomial defects at $(2, 3)$, generate only $\gcd(M - 2, A - 3)$. Dynamic versions are subexponential by the S-unit gcd theorem. Additional polynomial identities cannot bridge an exponential support gap unless they retain new prime-dependent order data.

The most promising overlooked direction is therefore neither another interpolation determinant nor another generic transcendence estimate. It is a local–global regulator or sparse-monoid interpolation theorem that converts the exact real logarithmic determinant into a non-generic supply of finite common exponents. The acceptance threshold is explicit: beat $P_3(q, e)$ and $P_4(q, e)$ at one depth, or retain any positive density through unbounded depth. That is a sharply falsifiable research objective and a tractable target for formal proof engineering.

A Derivation checks for the coherence numerators

For reference, the rank-four numerator may be rearranged as

$$\begin{aligned} N_4(q,e) &= 1 + \sum_{t=1}^{e} (q^{2t} - q^{2t-2}) q^t \\ &= 1 + (q^2 - 1) \sum_{t=1}^{e} q^{3t-2} \\ &= 1 + q(q^2-1) \frac{q^{3e}-1}{q^3-1}. \end{aligned}$$

Its leading contribution comes from primitive base vectors ($s = 0$, $t = e$):

$$(q^{2e} - q^{2e-2}) q^e = (1 - q^{-2}) q^{3e}.$$

After division by q^{4e} this gives

$$(1 - q^{-2}) q^{-e} \frac{1}{1 - q^{-3}} + O(q^{-4e}) = \frac{q(q+1)}{q^2+q+1} q^{-e} + O(q^{-4e}).$$

Similarly,

$$\begin{aligned} N_3(q,e) &= 1 + \sum_{t=1}^{e} (q^{2t} - q^{2t-1}) \\ &= 1 + q(q-1) \frac{q^{2e}-1}{q^2-1} \\ &= 1 + \frac{q(q^{2e}-1)}{q+1}, \end{aligned}$$

so

$$P_3(q,e) = \frac{q}{q+1} q^{-e} + O(q^{-3e}).$$

B A compact reproducibility script

The following script reproduces the pairwise, status-pattern, and coherence experiments. It uses only Python's exact integers. The depth-25 loop is intentionally separated because it uses a larger prime bound.

```python
from __future__ import annotations
from collections import Counter
from math import isqrt


def primes_upto(n: int) -> list[int]:
    sieve = bytearray(b"\x01") * (n + 1)
    sieve[:2] = b"\x00\x00"
    for p in range(2, isqrt(n) + 1):
        if sieve[p]:
            sieve[p*p:n+1:p] = b"\x00" * (((n - p*p)//p) + 1)
    return [i for i, flag in enumerate(sieve) if flag]


def residue_logs(vals: tuple[int, ...], p: int, Q: int, q: int):
    n = (p - 1)//Q
    omega = None
```

```
    for g in range(2, 200):
        candidate = pow(g, n, p)
        if pow(candidate, Q//q, p) != 1: # exact order Q
            omega = candidate
            break
    if omega is None:
        raise RuntimeError(("no root", p, Q))
    table = {}
    cur = 1
    for j in range(Q):
        table[cur] = j
        cur = cur * omega % p
    return tuple(table[pow(v % p, n, p)] for v in vals)


def coherent(t: tuple[int, int, int, int], Q: int) -> bool:
    u, v, m, a = t
    return any((k*u - m) % Q == 0 and (k*v - a) % Q == 0
               for k in range(Q))


def predicted(q: int, e: int, rank: int) -> float:
    Q = q**e
    if rank == 4:
        numerator = 1 + q*(q*q - 1)*(q**(3*e) - 1)//(q**3 - 1)
        return numerator/Q**4
    numerator = 1 + q*(q**(2*e) - 1)//(q + 1)
    return numerator/Q**3


def coherence_run(limit: int, q: int, e: int):
    Q = q**e
    primes = primes_upto(limit)
    examples = [
        ("rank4", (2, 3, 7, 11), 4),
        ("rank3", (2, 3, 7, 42), 3),
    ]
    for name, vals, rank in examples:
        total = good = relation_bad = 0
        for p in primes:
            if p % Q != 1 or any(v % p == 0 for v in vals):
                continue
            t = residue_logs(vals, p, Q, q)
            total += 1
            good += coherent(t, Q)
            if rank == 3 and (t[0] + t[1] + t[2] - t[3]) % Q:
                relation_bad += 1
        print(name, "Q", Q, "total", total, "good", good,
              "frequency", good/total,
              "predicted", predicted(q, e, rank),
              "relation_bad", relation_bad)


def is_qth_power(a: int, p: int, q: int) -> bool:
    return pow(a % p, (p - 1)//q, p) == 1


def status_run(limit: int = 5_000_000, q: int = 5):
    primes = primes_upto(limit)

    pair = Counter()
    for p in primes:
        if p % q == 1 and p not in (2, 7):
            pair[(is_qth_power(2, p, q), is_qth_power(14, p, q))] += 1
    print("pair", pair)

    for name, vals in [("rank4", (2, 3, 7, 11)),
                       ("rank3", (2, 3, 7, 42))]:
```

```
        counts = Counter()
        for p in primes:
            if p % q != 1 or any(v % p == 0 for v in vals):
                continue
            pattern = tuple(int(is_qth_power(v, p, q)) for v in vals)
            counts[pattern] += 1
        print(name, counts)


if __name__ == "__main__":
    status_run()
    coherence_run(5_000_000, 5, 1)
    coherence_run(10_000_000, 5, 2)
```

C Source and novelty audit

The nonduplication audit used the following checks on the supplied source.

- Section headings and labels were extracted programmatically; the source contains a unified report, two large continuations, twenty-seven shorter reports, and multiple appendices.
- Exact searches were run for the terminology central to this continuation. No hits were found for the support theorem authors, Chebotarev, the support problem, discrete logarithms, or congruence preservation. A normalized 20-token shingle comparison found no substantive common passage; its sole hit was shared theorem-environment boilerplate in the preambles.
- Generic Kummer theory, valuation lattices, and the Bugeaud–Corvaja–Zannier theorem do occur in the source. The present use differs: Kummer theory supplies exact residue-vector densities; the gcd theorem proves the dynamic additive-anchor ceiling.
- The additive classification, defect ideal, prime-power coherence counts, relation-pattern law, and local cyclic common-exponent criterion were derived independently for this report.

D Bibliography and public artifacts

References

[1] L. Alaoglu and P. Erdős, *On highly composite and similar numbers*, Transactions of the American Mathematical Society **56** (1944), 448–469. doi:10.1090/S0002-9947-1944-0011087-2.

[2] C. Corrales-Rodrigáñez and R. Schoof, *The support problem and its elliptic analogue*, Journal of Number Theory **64** (1997), 276–290. doi:10.1006/jnth.1997.2114.

[3] M. Bhargava, *P-orderings and polynomial functions on arbitrary subsets of Dedekind rings*, Journal für die reine und angewandte Mathematik **490** (1997), 101–128. doi:10.1515/crll.1997.490.101.

[4] Y. Bugeaud, P. Corvaja, and U. Zannier, *An upper bound for the G.C.D. of $a^n - 1$ and $b^n - 1$*, Mathematische Zeitschrift **243** (2003), 79–84. doi:10.1007/s00209-002-0449-z.

[5] P. Corvaja and U. Zannier, *A lower bound for the height of a rational function at S-unit points*, Monatshefte für Mathematik **144** (2005), 203–224; preprint 2003. doi:10.1007/s00605-004-0273-0.

[6] A. Perucca, P. Sgobba, and S. Tronto, *Explicit Kummer theory for the rational numbers*, International Journal of Number Theory **16** (2020), no. 10, 2213–2231. doi:10.1142/S1793042120501146.

[7] A. Perucca, P. Sgobba, and S. Tronto, *Kummer theory for number fields via entanglement groups*, Manuscripta Mathematica **169** (2022), 251–270. doi:10.1007/s00229-021-01328-0.

[8] A. Perucca, *On the reduction of points on abelian varieties and tori*, International Mathematics Research Notices **2011** (2011), 293–308. doi:10.1093/imrn/rnq080.

[9] C. Debry and A. Perucca, *Reductions of subgroups of the multiplicative group*, arXiv:1312.6620 (2013).

[10] P. Bruin and A. Perucca, *Reductions of points on algebraic groups, II*, arXiv:1802.08527 (2018).

[11] A. O. Gelfond, *Transcendental and Algebraic Numbers*, Dover, 1960; and Th. Schneider, *Einführung in die transzendenten Zahlen*, Springer, 1957.

[12] J. Neukirch, *Algebraic Number Theory*, Springer, 1999, Chapter VII.

[13] P. Nogueira Grossi, *An Independent Lean 4 Verification of the Alpöge–Fable Counterexample*, Zenodo, version 1.0, 23 July 2026. `https://zenodo.org/records/21514514`.

[14] S. Gao, *Counterexamples to the Jacobian conjecture in dimensions greater than two*, arXiv:2608.00222 (2026).

[15] NSF Institute for Computer-Aided Reasoning in Mathematics, *Elliptic Curve Rank Leaderboard, curve 273*, accessed 22 August 2026. `https://elliptic-rank.icarm.cloud/curve/273`.

[16] OEIS Foundation, *A007299 and A019442: Hadamard-matrix existence records*, updates of 12–13 August 2026. `https://oeis.org/A007299`; `https://oeis.org/A019442`.